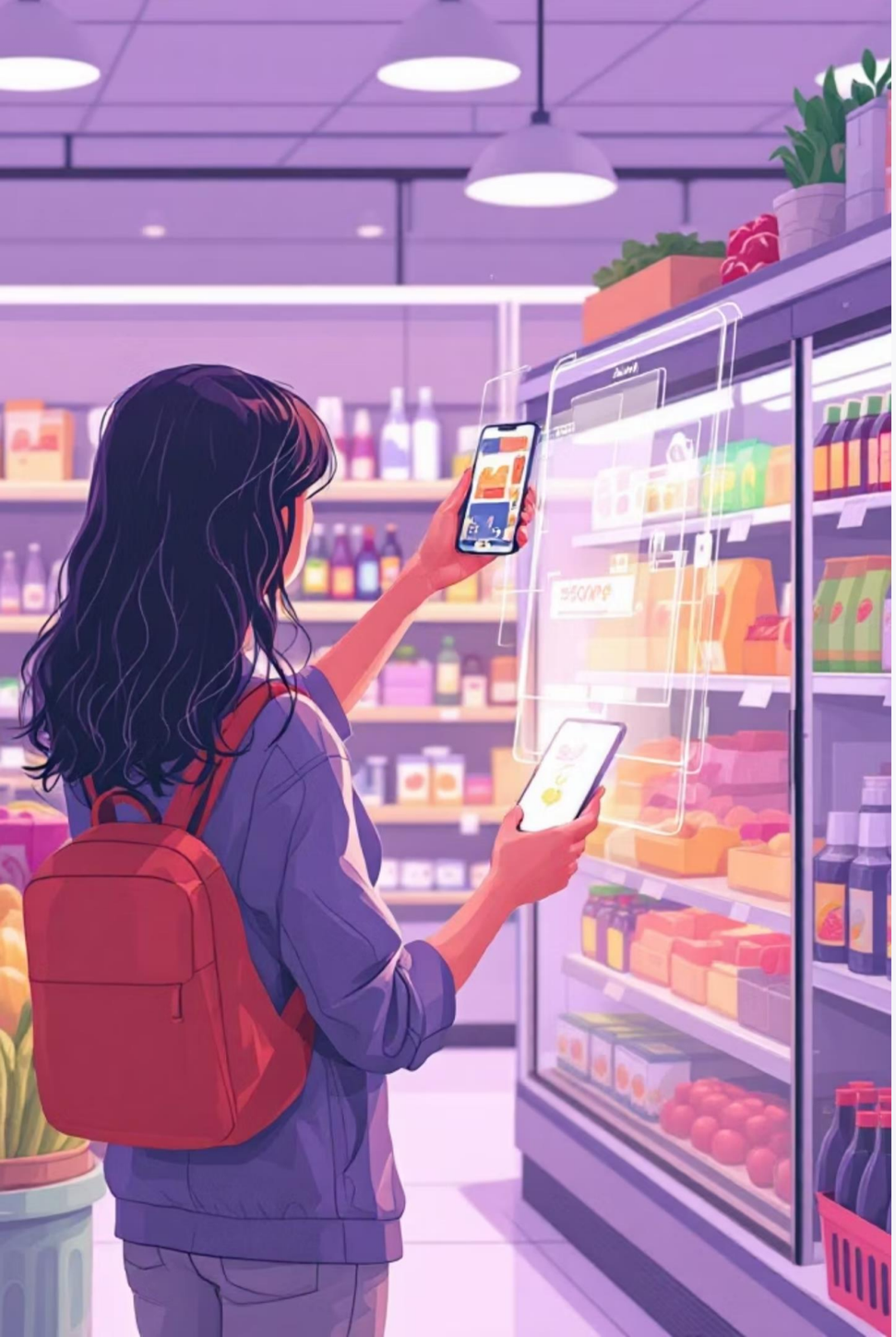

SafeCart: Real-Time Multimodal AI for Allergen-Free Grocery Shopping

A unified system combining computer vision, multilingual NLP, and augmented reality to enable safe, efficient grocery shopping for individuals with food allergies and dietary restrictions.

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DA 627 Project Proposal

The Global Food Allergy Challenge

200M+

People Affected

Estimated global population managing food allergies or dietary restrictions

30-60

Minutes Per Trip

Average time spent reading labels individually by allergy-sensitive shoppers

10.8%

Adult Prevalence

Percentage of US adults with at least one diagnosed food allergy

Grocery shopping remains hazardous due to unreadable foreign-language labels, inconsistent global allergen disclosure laws, and hidden ingredient synonyms across jurisdictions.

SafeCart: Five Modalities, One Unified System

01

Computer Vision

Product detection and shelf scanning via smartphone camera

02

OCR Extraction

Label text extraction with multilingual script support

03

Multilingual NLP

Ingredient parsing, translation, and allergen entity recognition

04

Database Cross-Reference

Structured allergen database with synonym resolution

05

AR Guidance

Real-time visual safety overlays and personalized alerts

How SafeCart Works

Real-Time Detection

Users point their smartphone at store shelves and instantly receive color-coded safety overlays in their native language.

Personalized Profiles

Individual allergen severity levels, cultural/religious restrictions, and nutritional goals mapped to real-time recommendations.

Smart List Generation

Automatically generates personalized, allergen-safe weekly shopping lists based on user profiles.

Technical Architecture

Image Capture & OCR

YOLO/EfficientDet detects products; Tesseract OCR extracts ingredient text with multilingual support for Arabic, Cyrillic, and CJK scripts

Multilingual NLP

XLM-RoBERTa fine-tuned on CC-100 corpus performs simultaneous translation and allergen entity recognition with custom synonym dictionary

Database Cross-Reference

OpenFoodFacts and USDA FoodData Central lookup with manufacturing metadata for cross-contamination detection

AR Overlay

ARCore/ARKit renders color-coded safety verdicts (green/amber/red) with ingredient breakdowns in preferred language

Dataset Foundation

OpenFoodFacts

2M+ products from 150+ countries with multilingual ingredients, allergen declarations, and product imagery

Food-101 & VIREO-172

211K food images spanning 273 categories for visual recognition training and evaluation

CC-100 / mC4

Large-scale multilingual web corpus spanning 100+ languages for domain-specific NLP fine-tuning

Model Evaluation Strategy

1

Precision & Recall

Measured on annotated ingredient lists from OpenFoodFacts test sets

2

F1-Score

Harmonic mean of precision and recall for NER evaluation

3

Language Coverage

Percentage of supported scripts achieving >85% F1-score

4

False Negative Rate

Safety-critical metric measuring proportion of actual allergens missed

5

Task Completion Time

Time required to identify safe products vs. manual baseline

Model Comparison: XLM-RoBERTa, mBERT, and FoodBERT evaluated for allergen NER performance

Expected Outcomes

Research Deliverables

- Comparative evaluation of multilingual NLP models for food allergen NER
- Benchmarked end-to-end pipeline performance
- Cross-contamination detection accuracy analysis

Technical Deliverables

- Documented source code repository
- Fine-tuned models with serialized weights
- FastAPI backend endpoints
- Functional mobile AR prototype

Broader Impact

Academic Relevance

Direct application of vision-language grounding, cross-lingual representation learning, and real-time multimodal inference from course topics

Professional Value

Develops competencies in multimodal integration, multilingual NLP, real-time edge inference, and mobile AR—2026's most in-demand AI skills

Societal Impact

Reduces anaphylaxis hospitalizations from accidental exposure while improving food autonomy for elderly, disabled, and international travelers

THANK YOU!!

Any Questions ?